

Probability Theory

Machine Learning (Bilingual)

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Probability Theory

Binary Variables

Multinomial Variables

The Gaussian Distribution

Bayes' Rule

Probability Theory



Random variables

A random variable is a variable that can take on different values randomly.

Probability distributions

A probability distribution is a description of how likely a random variable or set of random variables is to take on each of its possible states.



- A probability distribution over **discrete variables** may be described using a **probability mass function** (PMF).
- $P(X)$
- $P(X) = P(X = x)$ or $x \sim P(X)$
- Joint probability distribution, $P(x, y) = P(X = x, Y = y)$

Properties of PMF

- The domain of P must be the set of all possible states of X .
- $\forall x \in X, 0 \leq P(x) \leq 1$
- $\sum_{x \in X} P(x) = 1$



- Continuous variables
- The domain of p must be the set of all possible states of X .
- $\forall x \in X, p(x) \geq 0$
- $\int p(x)dx = 1$



The probability distribution over a *subset* is known as the **marginal** probability distribution.

- Suppose we have discrete random variables X and Y , and we know $P(X, Y)$. We can find $P(X)$ with the **sum rule**:

$$\forall x \in X, P(X = x) = \sum_y P(X = x, Y = y)$$

- For continuous variables, we need to use integration instead of summation:

$$p(x) = \int p(x, y) dy$$



- We are interested in the probability of some event, given that some other event has happened.
- We denote the **conditional probability** that $Y = y$ given $X = x$ as $P(Y = y|X = x)$.
- This is computed with:

$$P(Y = y|X = x) = P(Y = y, X = x)/P(X = x)$$



- Any joint probability distribution over many random variables may be decomposed into conditional distributions over only one variable:

$$P(X_1, \dots, X_n) = P(X_1) \prod_{i=1}^n P(X_i | X_1, \dots, X_{i-1})$$

- This observation is known as the **chain rule** or **product rule** of probability.

Examples

$$P(a, b, c) = P(a|b, c)P(b, c) = P(a|b, c)P(b|c)P(c)$$

$$P(a, b|c) = P(a|b, c)P(b|c)$$



- Two random variables X and Y are **independent** if their probability distribution can be expressed as a product of two factors, one involving only X and one involving only Y :

$$\forall x \in X, y \in Y$$

$$p(X = x, Y = y) = p(X = x)p(Y = y)$$



- Two random variables X and Y are **conditionally independent** given a random variable z if the conditional probability distribution over X and Y factorizes in this way for every value of Z :

$$\forall x \in X, y \in Y, z \in Z$$

$$p(X=x, Y=y|Z=z) = p(X=x|Z=z)p(Y=y|Z=z)$$



The expectation or expected value of function $f(x)$ w.r.t. a probability distribution $P(X)$ is the average or mean value that f takes on when x is drawn from P .

- For discrete variables:

$$\mathbb{E}_{x \sim P}[f(x)] = \sum_x P(x)f(x)$$

- For continuous variables:

$$\mathbb{E}_{x \sim P}[f(x)] = \int p(x)f(x)dx$$

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- For continuous variables:

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- Expectations are linear:

$$\mathbb{E}[\alpha f(x) + \beta g(x)] = \alpha \mathbb{E}[f(x)] + \beta \mathbb{E}[g(x)]$$



The variance gives a measure of how much the different values of a function are spread apart:

$$\text{Var}(f(x)) = \mathbb{E}\left[\left(f(x) - \mathbb{E}[f(x)]\right)^2\right]$$

- The square root of the variance is known as the **standard deviation**.



The covariance gives some sense of how much two values are linearly related to each other:

$$\text{Cov}(f(x), g(y)) = \mathbb{E}\left[(f(x) - \mathbb{E}[f(x)])(g(x) - \mathbb{E}[g(x)])\right]$$

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The covariance matrix of a random vector $\mathbf{x} \in \mathbb{R}^n$ is an $n \times n$ matrix:

$$\mathbf{x} \in \mathbb{R}^n, \text{Cov}(\mathbf{x})_{i,j} = \text{Cov}(x_i, x_j)$$

Binary Variables

- Coin flipping: heads=1, tails=0

$$p(x = 1|\mu) = \mu$$

- **Bernoulli** Distribution

$$\text{Bern}(x|\mu) = \mu^x (1 - \mu)^{1-x}$$

$$\mathbb{E}[x] = \mu$$

$$\text{var}[x] = \mu(1 - \mu)$$

- N coin flips

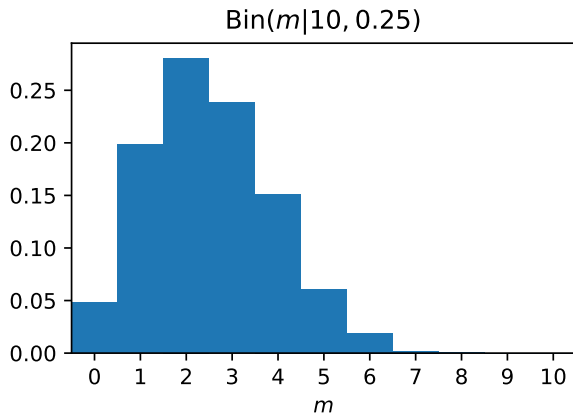
$$p(m \text{ heads} | N, \mu)$$

- Binomial Distribution

$$\text{Bin}(m|N, \mu) = \binom{N}{m} \mu^m (1 - \mu)^{N-m}$$

$$\mathbb{E}[m] = \sum_{m=0}^N \text{Bin}(m|N, \mu) = N\mu$$

$$\begin{aligned} \text{var}[m] &= \sum_{m=0}^N (m - E[m])^2 \text{Bin}(m|N, \mu) \\ &= N\mu(1 - \mu) \end{aligned}$$



Maximum likelihood (ML) for Bernoulli

- Given $\mathcal{D} = \{x_1, \dots, x_N\}$, m heads (1), $N - m$ tails (0)

$$p(\mathcal{D}|\mu) = \prod_{n=1}^N p(x_n|\mu) = \prod_{n=1}^N \mu^{x_n} (1 - \mu)^{1-x_n}$$

$$\begin{aligned} \ln p(\mathcal{D}|\mu) &= \sum_{n=1}^N \ln p(x_n|\mu) \\ &= \sum_{n=1}^N \{x_n \ln \mu + (1 - x_n) \ln(1 - \mu)\} \end{aligned}$$

$$\mu_{\text{ML}} = \frac{1}{N} \sum_{n=1}^N x_n = \frac{m}{N}$$



Example:

$$\mathcal{D} = \{1, 1, 1\} \rightarrow \mu_{ML} = \frac{3}{3} = 1$$

Prediction: **all** future tosses will land heads up.

Overfitting to $\mathcal{D}$

Distribution over $\mu \in [0, 1]$.

$$\text{Beta}(\mu|a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \mu^{a-1} (1-\mu)^{b-1}$$

$$\mathbb{E}[\mu] = \frac{a}{a+b}$$

$$\text{var}[\mu] = \frac{ab}{(a+b)^2(a+b+1)}$$



$$\Gamma(t) = \int_0^{\infty} x^{t-1} e^{-x} dx$$

Properties

$$\Gamma(t+1) = t\Gamma(t)$$

With $\Gamma(1) = 1$,

$$\Gamma(n) = 1 \cdot 2 \cdot 3 \cdots (n-1) = (n-1)!$$

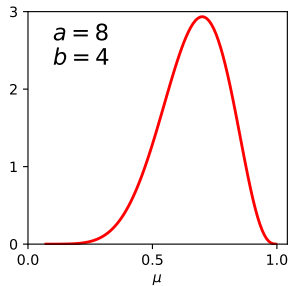
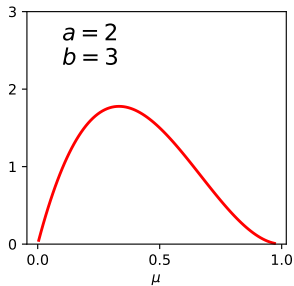
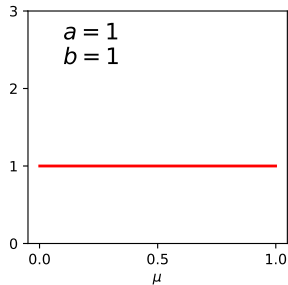
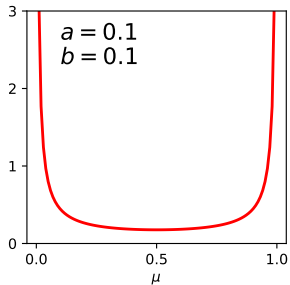


$$\begin{aligned} p(\mu|a_0, b_0, \mathcal{D}) &\propto p(\mathcal{D}|\mu)p(\mu|a_0, b_0) \\ &= \left(\prod_{n=1}^N \mu^{x_n} (1 - \mu)^{1-x_n} \right) \text{Beta}(\mu|a_0, b_0) \\ &\propto \mu^{m+a_0-1} (1 - \mu)^{(N-m)+b_0-1} \\ &\propto \text{Beta}(\mu|a_N, b_N) \end{aligned}$$

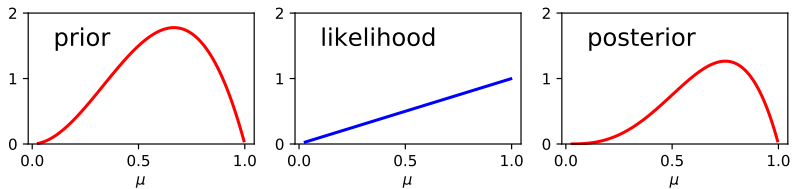
$$a_N = a_0 + m, \quad b_N = b_0 + (N - m)$$

The Beta distribution provides the **conjugate** prior for the Bernoulli distribution.

Beta Distribution



Prior · Likelihood = Posterior



As the size of the data set, N , increase

$$a_N \rightarrow m$$

$$b_N \rightarrow N - m$$

$$\mathbb{E}[\mu] = \frac{a_N}{a_N + b_N} \rightarrow \frac{m}{N} = \mu_{ML}$$

$$\text{var}\mu = \frac{a_N b_N}{(a_N + b_N)^2 (a_N + b_N + 1)} \rightarrow 0$$



What is the probability that the next coin toss will land heads up?

$$\begin{aligned} p(x = 1|a_0, b_0, \mathcal{D}) &= \int_0^1 p(x = 1|\mu)p(\mu|a_0, b_0, \mathcal{D})d\mu \\ &= \int_0^1 \mu p(\mu|a_0, b_0, \mathcal{D})d\mu \\ &= \mathbb{E}[\mu|a_0, b_0, \mathcal{D}] = \frac{a_N}{b_N} \end{aligned}$$

Multinomial Variables

1-of- K coding scheme:

$$\mathbf{x} = (0, 0, 1, 0, 0, 0)^T$$

$$p(\mathbf{x}|\boldsymbol{\mu}) = \prod_{k=1}^K \mu_k^{x_k}, \quad \forall k : \mu_k \geq 0 \text{ and } \sum_{k=1}^K \mu_k = 1$$

$$\mathbb{E}[\mathbf{x}|\boldsymbol{\mu}] = \sum_{\mathbf{x}} p(\mathbf{x}|\boldsymbol{\mu}) \mathbf{x} = (\mu_1, \dots, \mu_K)^T = \boldsymbol{\mu}$$

$$\sum_{\mathbf{x}} p(\mathbf{x}|\boldsymbol{\mu}) = \sum_{k=1}^K \mu_k = 1$$

Given: $\mathcal{D} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$

$$p(\mathcal{D}|\boldsymbol{\mu}) = \prod_{n=1}^N \prod_{k=1}^K \mu_k^{x_{nk}} = \prod_{k=1}^K \mu_k^{\sum_n x_{nk}} = \prod_{k=1}^K \mu_k^{m_k}$$

Ensure $\sum_k \mu_k = 1$, use a Lagrange multiplier, λ .

$$\sum_{k=1}^K m_k \ln \mu_k + \lambda \left(\sum_{k=1}^K \mu_k - 1 \right)$$

$$\mu_k = -m_k/\lambda \quad \mu_k^{\text{ML}} = \frac{m_k}{N}$$

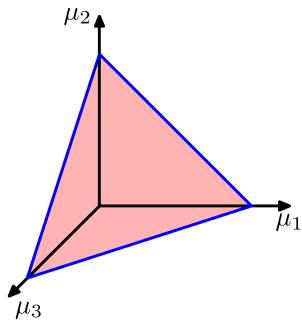


$$\text{Mult}(m_1, m_2, \dots, m_K | \boldsymbol{\mu}, N) = \binom{N}{m_1 m_2 \dots m_K} \prod_{k=1}^K \mu_k^{m_k}$$

$$\mathbb{E}[m_k] = N\mu_k$$

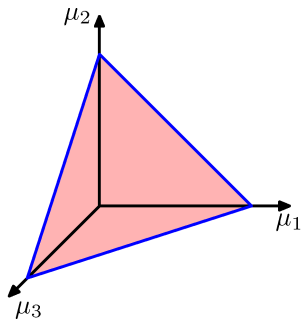
$$\text{var}[m_k] = N\mu_k(1 - \mu_k)$$

$$\text{cov}[m_j m_k] = -N\mu_j \mu_k$$



Prior distribution for parameter $\{\mu_k\}$:

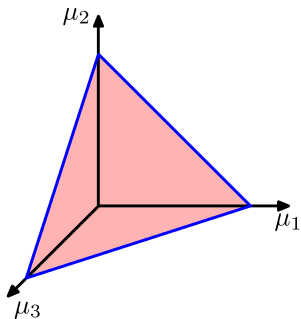
$$p(\boldsymbol{\mu}|\boldsymbol{\alpha}) \propto \prod_{k=1}^K \mu_k^{\alpha_k - 1}$$



Prior distribution for parameter $\{\mu_k\}$:

$$p(\boldsymbol{\mu}|\boldsymbol{\alpha}) \propto \prod_{k=1}^K \mu_k^{\alpha_k-1}$$

$$\text{Dir}(\boldsymbol{\mu}|\boldsymbol{\alpha}) = \frac{\Gamma(\alpha_0)}{\Gamma(\alpha_1) \cdots \Gamma(\alpha_K)} \prod_{k=1}^K \mu_k^{\alpha_k-1}$$

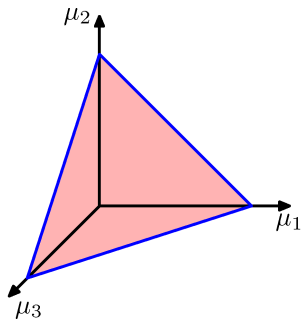


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$$\alpha_0 = \sum_{k=1}^K \alpha_k$$



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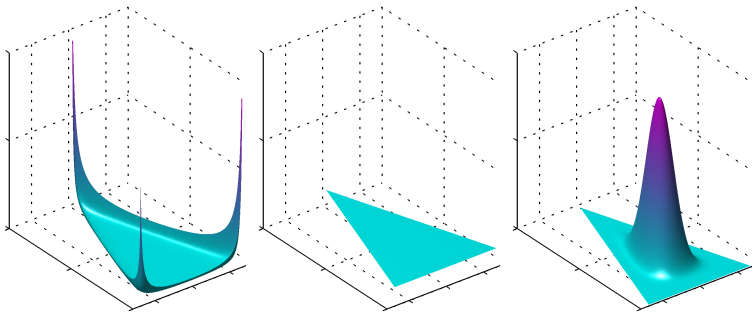
Conjugate prior for the multinomial distribution.



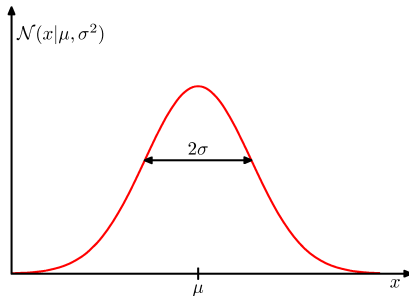
$$p(\boldsymbol{\mu}|\mathcal{D}, \boldsymbol{\alpha}) \propto p(\mathcal{D}|\boldsymbol{\mu})p(\boldsymbol{\mu}|\boldsymbol{\alpha}) \propto \prod_{k=1}^K \mu_k^{\alpha_k + m_k - 1}$$

$$\begin{aligned} p(\boldsymbol{\mu}|\mathcal{D}, \boldsymbol{\alpha}) &= \text{Dir}(\boldsymbol{\mu}|\boldsymbol{\alpha} + \boldsymbol{m}) \\ &= \frac{\Gamma(\alpha_0 + N)}{\Gamma(\alpha_1 + m_1) \cdots \Gamma(\alpha_K + m_K)} \prod_{k=1}^K \mu_k^{\alpha_k + m_k - 1} \end{aligned}$$

From left to right, α_k equals 10^{-1} , 10^0 , and 10^1 , respectively.



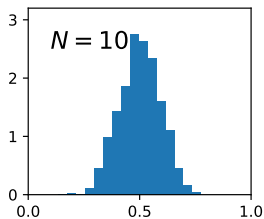
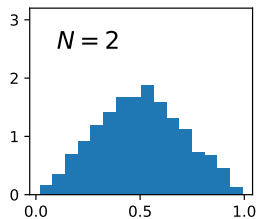
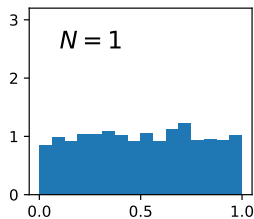
The Gaussian Distribution

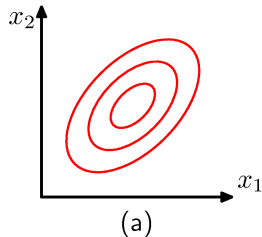


$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{1}{2\sigma^2} (x - \mu)^2 \right\}$$

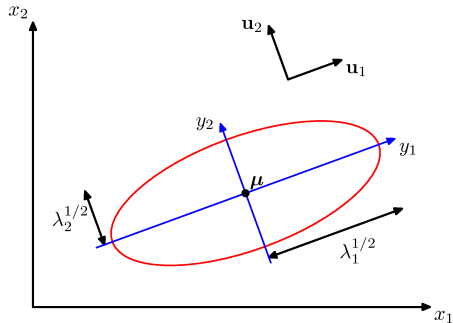
The distribution of the sum of N i.i.d. random variables becomes **increasingly Gaussian** as N grows.

Example: N uniform $[0, 1]$ random variables.





$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{\frac{D}{2}}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}$$



$$\Delta^2 = (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})$$

$$\boldsymbol{\Sigma}^{-1} = \sum_{i=1}^D \frac{1}{\lambda_i} \mathbf{u}_i \mathbf{u}_i^T$$

$$\Delta^2 = \sum_{i=1}^D \frac{y_i^2}{\lambda_i}$$

$$y_i = \mathbf{u}_i^T (\mathbf{x} - \boldsymbol{\mu})$$



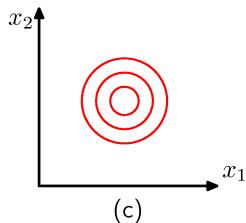
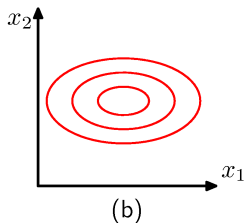
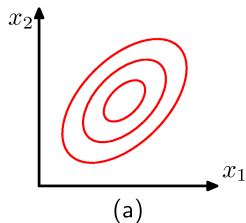
$$\begin{aligned}\mathbb{E}[\mathbf{x}] &= \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma|^{1/2}} \int \exp \left\{ -\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu}) \right\} \mathbf{x} d\mathbf{x} \\ &\stackrel{\underline{\underline{\mathbf{z}=\mathbf{x}+\boldsymbol{\mu}}}}{=} \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma|^{1/2}} \int \exp \left\{ -\frac{1}{2}\mathbf{z}^T \Sigma^{-1} \mathbf{z} \right\} (\mathbf{z}+\boldsymbol{\mu}) d\mathbf{z}\end{aligned}$$

Thanks to anti-symmetry of $\mathbf{z}$

$$\mathbb{E}[\mathbf{x}] = \boldsymbol{\mu}$$

$$\mathbb{E}[\mathbf{x}\mathbf{x}^T] = \boldsymbol{\mu}\boldsymbol{\mu}^T + \boldsymbol{\Sigma}$$

$$\text{cov}[\mathbf{x}] = \mathbb{E}[(\mathbf{x} - \mathbb{E}[\mathbf{x}])(\mathbf{x} - \mathbb{E}[\mathbf{x}])^T] = \boldsymbol{\Sigma}$$





$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

$$\mathbf{x} = \begin{pmatrix} \mathbf{x}_a \\ \mathbf{x}_b \end{pmatrix} \quad \boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_a \\ \boldsymbol{\mu}_b \end{pmatrix} \quad \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{aa} & \boldsymbol{\Sigma}_{ab} \\ \boldsymbol{\Sigma}_{ba} & \boldsymbol{\Sigma}_{bb} \end{pmatrix}$$

$$\boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1} \quad \boldsymbol{\Lambda} = \begin{pmatrix} \boldsymbol{\Lambda}_{aa} & \boldsymbol{\Lambda}_{ab} \\ \boldsymbol{\Lambda}_{ba} & \boldsymbol{\Lambda}_{bb} \end{pmatrix}$$

Partitioned conditionals

$$p(\mathbf{x}_a|\mathbf{x}_b) = \mathcal{N}(\mathbf{x}_a|\boldsymbol{\mu}_{a|b}, \boldsymbol{\Sigma}_{a|b})$$

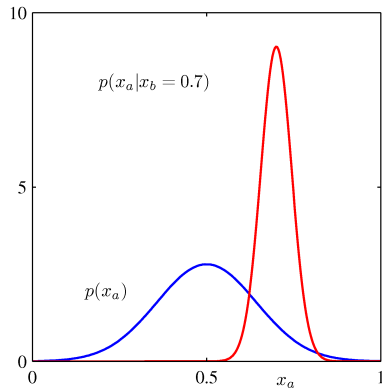
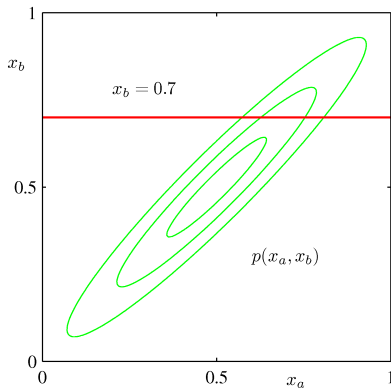
$$\boldsymbol{\Sigma}_{a|b} = \boldsymbol{\Lambda}_{aa}^{-1} = \boldsymbol{\Sigma}_{aa} - \boldsymbol{\Sigma}_{ab}\boldsymbol{\Sigma}_{bb}^{-1}\boldsymbol{\Sigma}_{ba}$$

$$\begin{aligned}\boldsymbol{\mu}_{a|b} &= \boldsymbol{\Sigma}_{a|b}\{\boldsymbol{\Lambda}_{aa}\boldsymbol{\mu}_a - \boldsymbol{\Lambda}_{ab}(\mathbf{x}_b - \boldsymbol{\mu}_b)\} \\ &= \boldsymbol{\mu}_a - \boldsymbol{\Lambda}_{aa}^{-1}\boldsymbol{\Lambda}_{ab}(\mathbf{x}_b - \boldsymbol{\mu}_b) \\ &= \boldsymbol{\mu}_a + \boldsymbol{\Sigma}_{ab}\boldsymbol{\Sigma}_{bb}^{-1}(\mathbf{x}_b - \boldsymbol{\mu}_b)\end{aligned}$$

Partitioned marginals

$$p(\mathbf{x}_a) = \int p(\mathbf{x}_a, \mathbf{x}_b)d\mathbf{x}_b = \mathcal{N}(\mathbf{x}_a|\boldsymbol{\mu}_a, \boldsymbol{\Sigma}_{aa})$$

Partitioned Conditionals and Marginals



Given i.i.d. data $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_N)^T$, the log likelihood function is given by

$$\begin{aligned}\ln p(\mathbf{X}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = & -\frac{ND}{2} \ln(2\pi) - \frac{N}{2} \ln |\boldsymbol{\Sigma}| \\ & - \frac{1}{2} \sum_{n=1}^N (\mathbf{x}_n - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}_n - \boldsymbol{\mu})\end{aligned}$$

Sufficient statistics

$$\sum_{n=1}^N \mathbf{x}_n \qquad \sum_{n=1}^N \mathbf{x}_n \mathbf{x}_n^T$$

Set the derivative of the log likelihood function to zero

$$\frac{\partial}{\partial \boldsymbol{\mu}} \ln p(\mathbf{X} | \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^N \boldsymbol{\Sigma}^{-1} (\mathbf{x}_n - \boldsymbol{\mu}) = 0$$

and solve to obtain

$$\boldsymbol{\mu}_{\text{ML}} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n$$

Similarly

$$\boldsymbol{\Sigma}_{\text{ML}} = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \boldsymbol{\mu}_{\text{ML}})(\mathbf{x}_n - \boldsymbol{\mu}_{\text{ML}})^T$$



Under the true distribution

$$\mathbb{E}[\boldsymbol{\mu}_{\text{ML}}] = \boldsymbol{\mu}$$

$$\mathbb{E}[\boldsymbol{\Sigma}_{\text{ML}}] = \frac{N-1}{N}\boldsymbol{\Sigma}$$

Hence define

$$\tilde{\boldsymbol{\Sigma}}_{\text{ML}} = \frac{1}{N-1} \sum_{n=1}^N (\mathbf{x}_n - \boldsymbol{\mu}_{\text{ML}})(\mathbf{x}_n - \boldsymbol{\mu}_{\text{ML}})^{\text{T}}$$

Bayes' Rule



- We know $P(y|x)$ and need to know $P(x|y)$. If we know $P(x)$, we can compute the desired quantity using **Bayes' rule**:

$$P(x|y) = \frac{P(x)P(y|x)}{P(y)}$$

- We do not need to begin with knowledge of $P(Y)$ since

$$P(y) = \sum_x P(y|x)P(x)$$



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Notes

- Named after *Reverend Thomas Bayes*, who first discovered a special case of the formula.
- The general version presented here was independently discovered by *Pierre-Simon Laplace*.

Given

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1})$$
$$p(\mathbf{y}|\mathbf{x}) = \mathcal{N}(\mathbf{y}|\mathbf{A}\mathbf{x} + \mathbf{b}, \mathbf{L}^{-1})$$

We have

$$p(\mathbf{y}) = \mathcal{N}(\mathbf{y}|\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{L}^{-1} + \mathbf{A}\boldsymbol{\Lambda}^{-1}\mathbf{A}^T)$$
$$p(\mathbf{x}|\mathbf{y}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\Sigma}\{\mathbf{A}^T\mathbf{L}(\mathbf{y} - \mathbf{b}) + \boldsymbol{\Lambda}\boldsymbol{\mu}\}, \boldsymbol{\Sigma})$$

where

$$\boldsymbol{\Sigma} = (\boldsymbol{\Lambda} + \mathbf{A}^T\mathbf{L}\mathbf{A})^{-1}$$